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Burns; John J. et al.

Refrigeration appliance with a door projection to reduce thermal transfer at door and cabinet interface

Abstract

A refrigeration appliance including: (a) a cabinet that includes a vacuum-insulated structure within which a refrigeration compartment configured to hold a food item is disposed; and (b) a door that includes a vacuum-insulated door structure coupled to the cabinet, the vacuum-insulated door including (i) a gasket that forms a seal against the vacuum-insulated cabinet when the door is in a closed position and (ii) a projection inward of the gasket that projects into the refrigeration compartment and is separated from the vacuum-insulated cabinet by a gap, the gap having a shortest distance that is less than 12 mm.

Inventors: Burns; John J. (Saint Joseph, MI), Karl; Jackson F. (Stevensville, MI), Lessa; Marcel (Joinville, BR), Rotter; Chad (Saugatuck, MI), Phalak; Sanket Vivek (Pune, IN)

Applicant: WHIRLPOOL CORPORATION (Benton Harbor, MI)

Family ID: 1000008780480

Assignee: Whirlpool Corporation (Benton Harbor, MI)

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2345743	12/1943	Gaston	N/A	N/A
3359053	12/1966	Hagendorn	N/A	N/A
4469383	12/1983	Losert	N/A	N/A
4653819	12/1986	Swerbinsky	N/A	N/A
4916864	12/1989	Thompson	N/A	N/A
5533311	12/1995	Tirrell	49/501	E06B 7/2312
6266970	12/2000	Nam et al.	N/A	N/A
8083305	12/2010	Lee et al.	N/A	N/A
10563905	12/2019	Allard	N/A	F25D 23/085
10753669	12/2019	Dherde	N/A	F25D 23/065
10807298	12/2019	Allard	N/A	F25D 23/064
10837697	12/2019	Amorim	N/A	B29C 45/0001
11320193	12/2021	Visin	N/A	F25D 23/085
11577446	12/2022	Allard	N/A	B29C 51/082
2003/0222553	12/2002	Lee	312/405	F25D 23/087
2009/0007587	12/2008	Lanzi et al.	N/A	N/A
2018/0311884	12/2017	Allard	N/A	B65D 81/3823
2019/0101320	12/2018	Dherde	N/A	F25D 23/069
2019/0128591	12/2018	Visin	N/A	F25D 23/02
2020/0132359	12/2019	Amorim	N/A	B29C 45/0046
2020/0406527	12/2019	Allard	N/A	B29C 51/02
2021/0207878	12/2020	Deka	N/A	F25D 23/085
2021/0348831	12/2020	Kapure	N/A	F25D 23/082
2022/0252333	12/2021	Awachat et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
725546	12/1954	GB	N/A
S613387	12/1985	JP	N/A
2012202582	12/2011	JP	N/A
20030094550	12/2002	KR	N/A

Primary Examiner: Wright; Kimberley S

Attorney, Agent or Firm: Price Heneveld LLP

Background/Summary

BACKGROUND OF THE DISCLOSURE

(1) The present disclosure generally relates to a refrigeration appliance, and more specifically, to a refrigeration appliance with a door with a projection that extends into a cabinet to reduce air flow and thus thermal transfer to and through an interface of the door and the cabinet.

(2) Refrigeration appliances are used to maintain a food item at a temperature below a temperature of an external environment (e.g., room temperature). Sometimes the refrigeration appliance includes a door that is attached to a cabinet, which forms a refrigeration compartment that holds the food item. In addition, sometimes the door of the refrigeration appliance includes a vacuum-insulated structure.

(3) However, there is a problem in that thermal transfer from the refrigeration compartment within the cabinet to the external environment across an interface between the door and the cabinet is suboptimally high. The greater the thermal transfer across the interface, the more the refrigeration appliance has to work to maintain the temperature of the refrigeration compartment (and the food item). The more the refrigeration appliance has to work, the more electricity the refrigeration appliance consumes.

SUMMARY OF THE DISCLOSURE

(4) The present disclosure addresses the aforementioned problem with a projection of the door that extends into a refrigeration compartment of the cabinet to form a narrow gap between the projection and the cabinet that reduces air flow toward, and thus thermal transfer through, the interface between the door and the cabinet.

(5) According to one aspect of the present disclosure, a refrigeration appliance comprises: (a) a cabinet comprising (i) a forward surface that is substantially vertical, (ii) interior surfaces defining a refrigeration compartment configured to hold a food item, and (iii) an opening into the refrigeration compartment disposed inward of the forward surface; and (b) a door coupled to the cabinet, the door comprising (i) a closed position that denies access through the opening and into the refrigeration compartment of the cabinet from an external environment, (ii) an open position that allows access through the opening to the refrigeration compartment from the external environment, the door being movable to, from, and between the closed position and the open position, (iii) an inner liner facing the refrigeration compartment when the door is in the closed position, (iv) an outer wrapper separated from the inner liner by a space, and (v) a trim breaker coupled to both the inner liner and the outer wrapper, the trim breaker comprising a projection that extends rearward into the refrigeration compartment of the cabinet, when the door is in the closed position, wherein, the projection of the trim breaker opposes one or more of the interior surfaces of the cabinet defining the refrigeration compartment, when the door is in the closed position, but is separated therefrom by a gap.

(6) According to another aspect of the present disclosure, a refrigeration appliance comprises: (a) a cabinet that comprises a vacuum-insulated structure within which a refrigeration compartment configured to hold a food item is disposed; and (b) a door that comprises a vacuum-insulated door structure coupled to the cabinet, the vacuum-insulated door comprising (i) a gasket that forms a seal against the vacuum-insulated cabinet when the door is in a closed position and (ii) a projection inward of the gasket that projects into the refrigeration compartment and is separated from the vacuum-insulated cabinet by a gap, the gap having a shortest distance that is less than 12 mm.

(7) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

- (2) FIG. 1 is a perspective view of a refrigeration appliance of the present disclosure built into cabinetry with a door in the closed position;
- (3) FIG. 2 is a perspective view of the refrigeration appliance of FIG. 1, illustrating the door in an open position to reveal a cabinet with a refrigeration compartment;
- (4) FIG. 3 is a perspective view of the refrigeration appliance of FIG. 1, illustrating the cabinet including a vacuum-insulated structure;
- (5) FIG. 4 is a front elevational view of the refrigeration appliance of FIG. 1;
- (6) FIG. 5 is an elevational view of a cross-section of the refrigeration appliance of FIG. 1 taken through line V-V of FIG. 4, illustrating the vacuum-insulated structure of the cabinet including an outer wrapper and an inner liner defining at least in part the interior surfaces of the refrigeration compartment;
- (7) FIG. 6 is a magnified view of area VI of FIG. 5, illustrating the door of the refrigeration appliance of FIG. 1 also including a vacuum-insulated structure with an outer wrapper, an inner liner, and a trim breaker coupled to both the outer wrapper and the inner liner;
- (8) FIG. 7 is the same view as FIG. 6 but from a slight perspective, illustrating the vacuum-insulated structure of the cabinet further including a trim breaker coupled to both the outer wrapper and the inner liner of the vacuum-insulated structure of the cabinet;
- (9) FIG. 8 is a magnified view of area VIII of FIG. 6, illustrating (i) the door further including a gasket forming a seal against a forward surface of the cabinet provided by the outer wrapper and (ii) the trim breaker of the vacuum-insulated structure of the door including a projection that extends rearward into the refrigeration compartment of the cabinet and separated from the trim breaker of the vacuum-insulated structure of the cabinet by a gap of relatively small proportions to reduce air flow from deeper within the refrigeration cabinet to the seal (e.g., interface) between the door and the cabinet that the gasket forms;
- (10) FIG. 9 is an elevation view of a cross-section of the refrigeration appliance of FIG. 1 taken through line IX-IX of FIG. 4, illustrating the inner liner of the vacuum-insulated structure of the cabinet providing at least a portion of the interior surfaces defining the refrigeration compartment;
- (11) FIG. 10 is a side elevation view of the refrigeration appliance of FIG. 1;
- (12) FIG. 11 is an elevation view of a cross-section of the refrigeration appliance of FIG. 1 taken through line XI-XI of FIG. 10, illustrating the inner liner of the vacuum-insulated structure of the cabinet providing the sidewalls, ceiling, and floor of the refrigeration compartment;
- (13) FIG. 12 is an elevation view of the trim breaker of the vacuum-insulated structure of the door; and
- (14) FIG. 13 is a magnified perspective view of area XIII of FIG. 12, illustrating the trim breaker of the vacuum-insulated structure of the door further including an outer perimeter, a rear recess inward of the outer perimeter, and the projection inward of the rear recess.
- (15) The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles described herein.

DETAILED DESCRIPTION

(16) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a refrigeration appliance. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further,

like numerals in the description and drawings represent like elements.

(17) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the disclosure as oriented in FIG. 1. Unless stated otherwise, the term “front” shall refer to the surface of the element closer to an intended viewer, and the term “rear” shall refer to the surface of the element further from the intended viewer. However, it is to be understood that the disclosure may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(18) The terms “including,” “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises a . . .” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(19) Referring to FIGS. 1 and 2, a refrigeration appliance **10** includes a cabinet **12** and a door **14**. The cabinet **12** includes a refrigeration compartment **16** and an opening **18** into the refrigeration compartment **16**. The refrigeration compartment **16** is configured to hold a food item **20**. The door **14** is coupled to the cabinet **12**, such as through hinges **22** that are attached to both the door **14** and the cabinet **12**. The hinges **22** can be articulating hinges. The door **14** is movable to, from, and between a closed position **24** (see, e.g., FIG. 1) and an open position **26** (see, e.g., FIG. 2). In the closed position **24**, the door **14** denies access through the opening **18** and into the refrigeration compartment **16** from an external environment **28**. In the open position **26**, the door **14** allows access through the opening **18** to the refrigeration compartment **16** from the external environment **28**. The refrigeration appliance **10** can be a freezer that maintains air and the food item **20** within the refrigeration compartment **16** at a temperature below a freezing point of water, such as within a range of from -18° C. to -23° C. The refrigeration appliance **10** can be a refrigerator that maintains the air and the food item **20** within the refrigeration compartment **16** at a temperature below room temperature but above the freezing point of water, such as within a range of from 2° C. to 4° C. The refrigeration appliance **10** can include multiple refrigeration compartments **16** that are maintained at difference temperatures (e.g., one of the refrigeration compartments **16** as the freezer and another one of the refrigeration compartments **16** as the refrigerator). The refrigeration appliance **10** can be of a built-in variety, where the cabinet **12** is designed to be recessed into cabinetry **30**, as illustrated. Alternatively, the refrigeration appliance **10** can be of a stand-alone variety, where the cabinet **12** is designed to be seen from the external environment **28** and thus may include a wrapper (not illustrated) over the cabinet **12**.

(20) Referring additionally to FIGS. 3-10, the cabinet **12** provides a forward surface **32**. The forward surface **32** faces forward **34**. The forward surface **32** faces the door **14** when the door **14** is in the closed position **24**. The forward surface **32** is substantially vertical. The opening **18** into the refrigeration compartment **16** is disposed laterally inward of the forward surface **32**. For example, the forward surface **32** frames the opening **18**. The door **14** forms a seal **36** against the forward surface **32** of the cabinet **12**, when the door **14** is in the closed position **24**, as further discussed below.

(21) The cabinet **12** includes interior surfaces **38** that define the refrigeration compartment **16**. For example, as in the illustrated embodiments, the interior surfaces **38** can be those provided by a rear wall **40**, side walls **42**, **44**, a ceiling **46**, and a floor **48** that collectively define the refrigeration compartment **16**. The ceiling **46** and the floor **48** oppose each other. The side walls **42**, **44** oppose each other. The rear wall **40** and the door **14** oppose each other, when the door **14** is in the closed

position **24**. The refrigeration compartment **16** is configured to hold the food item **20** in terms of the size of the refrigeration compartment **16** being large enough to hold the food item **20**. Further, the refrigeration compartment **16** can include one or more shelves **50** and the like that provide a horizontal surface **52** upon which the food item **20** can sit without falling to the floor **48** or out of the refrigeration compartment **16** when the door **14** is in the open position **26**.

(22) In embodiments, the cabinet **12** includes a vacuum-insulated structure **54**. The cabinet **12** then can include an inner liner **56**, an outer wrapper **58**, and a trim breaker **60**. The outer wrapper **58** at least partially envelops the inner liner **56**. For example, the outer wrapper **58** can include a top section **62** that is disposed above a top section **64** of the inner liner **56**, a rear section **66** that is rearward of a rear section **68** of the inner liner **56**, opposing side sections **70**, **72** that are disposed laterally outward of side sections **74**, **76** of the inner liner **56**, and so on. In such embodiments, the top section **64** of the inner liner **56** can form the ceiling **46** of the refrigeration compartment **16**, the rear section **68** of the inner liner **56** can form the rear wall **40** of the refrigeration compartment **16**, and the side sections **74**, **76** of the inner liner **56** can form the side walls **42**, **44** of the refrigeration compartment **16**. In addition to at least partially enveloping the inner liner **56**, the outer wrapper **58** of vacuum-insulated structure **54** can provide the forward surface **32** of the cabinet **12** that forms the seal **36** with the door **14**, when the door **14** is in the closed position **24**.

(23) The trim breaker **60** is coupled to both the inner liner **56** and the outer wrapper **58**. For example, as in the illustrated embodiments, the trim breaker **60** includes a forward recess **78** and a rear recess **80**. The forward recess **78** is disposed forward **34** relative to the rear recess **80**. An edge **82** of the outer wrapper **58** of the vacuum-insulated structure **54** of the cabinet **12** is disposed within the forward recess **78** of the trim breaker **60**, to couple the trim breaker **60** and the outer wrapper **58** together. Adhesive (not illustrated) can be utilized to affix the trim breaker **60** and the outer wrapper **58** in an air-tight manner. Similarly, an edge **84** of the inner liner **56** of the vacuum-insulated structure **54** of the cabinet **12** is disposed within the rear recess **80** of the trim breaker **60**, to couple the trim breaker **60** and the inner liner **56** together. Again, adhesive (not illustrated) can be utilized to affix the trim breaker **60** and the inner liner **56** in an air-tight manner.

(24) The outer wrapper **58** is separated from the inner liner **56** by a space **86**. The space **86** has an air pressure that is less than atmospheric pressure at the external environment **28**. The reduced pressure relative to the external environment **28** is a consequence of evacuating air from the space **86** after assembly of vacuum-insulated structure **54**. Reducing the amount of air within the space **86** reduces thermal transfer through the space **86**. The outer wrapper **58** and the inner liner **56** of the vacuum-insulated structure **54** of the cabinet **12** can be fabricated from sheet metal (either unitary or disparate panels welded together) or a plastic material. The trim breaker **60** can be a plastic material. Insulation material **87** (e.g., fumed silica) can occupy the space **86**.

(25) The door **14** includes a vacuum-insulated structure **88**. The vacuum-insulated structure **88** of the door **14** can include an inner liner **90** and an outer wrapper **92**. The inner liner **90** faces the refrigeration compartment **16**, more particularly the rear wall **40** thereof, when the door **14** is in the closed position **24**. The outer wrapper **92** faces the external environment **28** and is disposed forward **34** of the inner liner **90**, when the door **14** is in the closed position **24**. The outer wrapper **92** and the inner liner **90** are separated by a space **94**. Like the space **86** of the cabinet **12**, the space **94** of the door **14** has an air pressure that is less than atmospheric pressure at the external environment **28**. The reduced pressure relative to the external environment **28** is a consequence of evacuating air from the space **94** after assembly of vacuum-insulated structure **88**. Reducing the amount of air within the space **94** reduces thermal transfer through the space **94**. The inner liner **90** and the outer wrapper **92** can be formed of stamped sheet metal or a plastic material. Insulation material **96** (e.g., fumed silica) can occupy the space **94**.

(26) The vacuum-insulated structure **88** of the door **14** further includes a trim breaker **98**. The trim breaker **98** is coupled to both the inner liner **90** and the outer wrapper **92** of the vacuum-insulated structure **88** of the door **14**. For example, as in the illustrated embodiments, the trim breaker **98** of

the door **14** includes an outer forward recess **100** and an inner forward recess **102**. The outer forward recess **100** and the inner forward recess **102** are “forward” in the sense that they are open from the forward **34** direction (e.g., away from the refrigeration compartment **16**). The outer forward recess **100** is disposed closer to an outer perimeter **104** of the trim breaker **98** than the inner forward recess **102**. An edge **106** of the outer wrapper **92** of the door **14** is disposed within the outer forward recess **100** of the trim breaker **98**, to couple the trim breaker **98** and the outer wrapper **92** together. Adhesive **108** can be utilized to form an air-tight seal there. An edge **110** of the inner liner **90** of the door **14** is disposed within the inner forward recess **102** of the trim breaker **98**, to couple the trim breaker **98** and the inner liner **90** together. Likewise, adhesive (not illustrated) can be utilized to form an air-tight seal there. The trim breaker **98** can be formed of a plastic material.

(27) The door **14** further includes a gasket **114**. The gasket **114** forms the seal **36** against the forward surface **32** of the cabinet **12** when the door **14** is in the closed position **24**. In embodiments where the cabinet **12** includes the vacuum-insulated structure **54** with the outer wrapper **58** providing the forward surface **32**, the gasket **114** forms the seal **36** against the outer wrapper **58**. The gasket **114** can include a rearward portion **116** with a magnet **118**, a central portion **120**, and a forward portion **122**. The central portion **120** is disposed between the rearward portion **116** and the forward portion **122**. The gasket **114** can be made of a plastic, aside from the magnet **118**. The rearward portion **116** can provide a rear surface **124** that is substantially vertical and positioned to interface with and form the seal **36** against the forward surface **32** of the cabinet **12** (e.g., the outer wrapper **58** thereof), when the door **14** is in the closed position **24**. The magnet **118** draws the gasket **114** and thus the door **14** to the forward surface **32** of the cabinet **12** to seal **36** the door **14** to the cabinet **12**.

(28) The trim breaker **98** of the door **14** further includes a rear recess **126**. The rear recess **126** is open from a rearward direction **128**. The rear recess **126** is disposed closer to the outer perimeter **104** of the trim breaker **98** than the inner forward recess **102** of the trim breaker **98**. The rear recess **126** is disposed further from the outer perimeter **104** than the outer forward recess **100** of the trim breaker **98**. The forward portion **122** of the gasket **114** is disposed within the rear recess **126**, to couple the trim breaker **98** and the gasket **114** together. The forward portion **122** of the gasket **114** can friction fit into the rear recess **126** to attach the door **14** gasket **114** and the trim breaker **98** together.

(29) The trim breaker **98** of the door **14** further includes a projection **130**. The projection **130** extends rearward **128** relative to the inner forward recess **102** and the gasket **114** of the door **14**. The projection **130** extends rearward **128** through the opening **18** of the cabinet **12** and into the refrigeration compartment **16** when the door **14** is in the closed position **24**. The projection **130** opposes one or more of the interior surfaces **38** of the cabinet **12** that define the refrigeration compartment **16**, when the door **14** is in the closed position **24**. However, in embodiments, the projection **130** is separated from the interior surfaces **38** by a gap **132**. In other words, in those embodiments, the projection **130** does not contact the cabinet **12**. However, in other embodiments, the projection **130** may contact the cabinet **12**. The gap **132** being present can be advantageous because manufacturing tolerances can cause variations in the dimensions of the cabinet **12** and the door **14**. The gap **132** being present allows for such variations to exist without the projection **130** preventing the door **14** from moving to the closed position **24**.

(30) In embodiments, such as those illustrated, the trim breaker **60** of the cabinet **12** provides one or more of the interior surfaces **38** of the cabinet **12** that define the refrigeration compartment **16**. For example, the trim breaker **60** can form part of the ceiling **46** of the refrigeration compartment **16**, along with the inner liner **56**. In that instance, the trim breaker **60** forms a forward part **134** of the ceiling **46**, and the inner liner **56** may form a remainder **136** of the ceiling **46** of the refrigeration compartment **16**. When the door **14** is in the closed position **24**, the projection **130** of the trim breaker **98** of the door **14** opposes the trim breaker **60** of the cabinet **12**.

(31) The trim breaker **98** has a top section **138** where the projection **130** faces the ceiling **46** of the refrigeration compartment **16**. In embodiments, at the top section **138**, the trim breaker **98** has a top horizontal wall **142**. A top vertical wall **144** extends downward **146** from the top horizontal wall **142**. A second horizontal wall **148** extends forward **34** from the top vertical wall **144**. A second vertical wall **150** extends downward **146** from the second horizontal wall **148**. A third horizontal wall **152** extends rearward **128** from the second vertical wall **150**. A fourth horizontal wall **154** extends rearward **128** from the third horizontal wall **152**. The trim breaker **98** further includes a downward transition **156** from the third horizontal wall **152** to the fourth horizontal wall **154**. A third vertical wall **158** extends downward **146** from the fourth horizontal wall **154**. A fifth horizontal wall **160** extends forward **34** from the third vertical wall **158**. An angled wall **162** extends rearward **128** and downward **146** from the third vertical wall **158**. A fourth vertical wall **164** extends downward **146** from the angled wall **162**. A sixth horizontal wall **166** extends forward **34** from the fourth vertical wall **164**.

(32) The top horizontal wall **142**, the top vertical wall **144**, and the second horizontal wall **148** of the trim breaker **98** define the outer forward recess **100** of the trim breaker **98** of the door **14**. As mentioned, the edge **106** of the outer wrapper **92** of the door **14** is disposed within the outer forward recess **100** of the trim breaker **98** of the door **14**. More particularly, the edge **106** of the outer wrapper **92** is disposed horizontally between the top horizontal wall **142** and the second horizontal wall **148** of the trim breaker **98** and forward **34** of the top vertical wall **144** of the trim breaker **98**.

(33) The second horizontal wall **148**, the second vertical wall **150**, and the third horizontal wall **152** of the trim breaker **98** together define the rear recess **126** of the trim breaker **98** of the door **14**. As mentioned, the forward portion **122** of the gasket **114** of the door **14** is disposed within the rear recess **126** to couple the trim breaker **98** and the gasket **114** of the door **14** together. The forward portion **122** of the gasket **114** is disposed rearward **128** of the second vertical wall **150** and between the second horizontal wall **148** and the third horizontal wall **152**.

(34) The fourth horizontal wall **154**, the third vertical wall **158**, and the fifth horizontal wall **160** together define the inner forward recess **102** of the trim breaker **98** of the door **14**. As mentioned, the edge **110** of the inner liner **90** of the door **14** is disposed within the inner forward recess **102** to couple the inner liner **90** and the trim breaker **98** of the door **14** together. The edge **110** is disposed horizontally between the fourth horizontal wall **154** and the fifth horizontal wall **160**, and forward **34** of the third vertical wall **158**.

(35) The angled wall **162**, the fourth vertical wall **164**, and the sixth horizontal wall **166** of the trim breaker **98** together define the projection **130**. As mentioned, the projection **130** opposes one or more of the interior surfaces **38** of the cabinet **12** that define the refrigeration compartment **16**, when the door **14** is in the closed position **24**. More particularly, in embodiments, the angled wall **162** of the trim breaker **98** opposes one or more of the interior surfaces **38** of the cabinet **12** that define the refrigeration compartment **16**.

(36) The gap **132** separating the projection **130** of the trim breaker **98** of the door **14** from the refrigeration compartment **16** (e.g., the ceiling **46**, or the portion of the ceiling **46** provided by the trim breaker **60** of the cabinet **12**) has a shortest distance **168** between the projection **130** and the surface of the cabinet **12** defining the refrigeration compartment **16**, when the door **14** is in the closed position **24**. In embodiments, the shortest distance **168** is less than 12 mm. In embodiments, the shortest distance **168** is within a range of from 4 mm to 12 mm. In other embodiments, the shortest distance **168** is zero (e.g., the projection **130** contacts the surface of the cabinet **12** defining the refrigeration compartment **16**). In embodiments, the projection **130** extends rearward **128** at least 30 mm from a plane **170** that the forward surface **32** of the cabinet **12** forms.

(37) The refrigeration appliance **10** of the present disclosure addresses the problem set forth in the background of suboptimally high thermal transfer across the interface (e.g., where the seal **36** is formed) between the door **14** and the cabinet **12** (e.g., from the external environment **28** through

the interface and into the refrigeration compartment **16**) because the projection **130** of the trim breaker **98** of the door **14** extends into the refrigeration compartment **16** and is separated from the cabinet **12** (e.g., the trim breaker **60** of the vacuum-insulated structure **54** of the cabinet **12**) by the gap **132** with the shortest distance **168** that is relatively small. The gap **132** being relatively small hinders air flow from further within the refrigeration compartment **16**, through the gap **132**, and to the gasket **114** of the door **14** interfacing with the forward surface **32** of the cabinet **12**. Hindering the air flow hinders thermal transfer across that interface as well as the flow of relatively cool air across the interface and into the external environment **28**. The reduced thermal transfer results in less electricity that the refrigeration appliance **10** has to consume to maintain the temperature of the refrigeration compartment **16** and thus the food item **20**.

(38) According to a first aspect, a refrigeration appliance comprises: (a) a cabinet comprising (i) a forward surface that is substantially vertical, (ii) interior surfaces defining a refrigeration compartment configured to hold a food item, and (iii) an opening into the refrigeration compartment disposed inward of the forward surface; and (b) a door coupled to the cabinet, the door comprising (i) a closed position that denies access through the opening and into the refrigeration compartment of the cabinet from an external environment, (ii) an open position that allows access through the opening to the refrigeration compartment from the external environment, the door being movable to, from, and between the closed position and the open position, (iii) an inner liner facing the refrigeration compartment when the door is in the closed position, (iv) an outer wrapper separated from the inner liner by a space, and (v) a trim breaker coupled to both the inner liner and the outer wrapper, the trim breaker comprising a projection that extends rearward into the refrigeration compartment of the cabinet, when the door is in the closed position, wherein, the projection of the trim breaker opposes one or more of the interior surfaces of the cabinet defining the refrigeration compartment, when the door is in the closed position, but is separated therefrom by a gap.

(39) According to a second aspect of the present disclosure, the refrigeration appliance of the first aspect is presented, wherein the cabinet further comprises (i) an inner liner, (ii) an outer wrapper at least partially enveloping the inner liner and separated from the inner liner by a space, and (iii) a trim breaker coupled to both the inner liner and the outer wrapper.

(40) According to a third aspect of the present disclosure, the refrigeration appliance of the second aspect is presented, wherein (i) the outer wrapper provides the forward surface, and (ii) the door further comprises a gasket that forms a seal against the forward surface of the outer wrapper when the door is in the closed position.

(41) According to a fourth aspect of the present disclosure, the refrigeration appliance of any one of the second through third aspects is presented, wherein the space of the cabinet between the inner liner and the outer wrapper has an air pressure that is less than atmospheric pressure.

(42) According to a fifth aspect of the present disclosure, the refrigeration appliance of any one of the second through fourth aspects is presented, wherein the trim breaker of the cabinet comprises (i) a forward recess within which an edge of the outer wrapper of the cabinet is disposed to couple the trim breaker and the outer wrapper of the cabinet together and (ii) a rear recess within which an edge of the inner liner of the cabinet is disposed to couple the trim breaker and the inner liner of the cabinet together.

(43) According to a sixth aspect of the present disclosure, the refrigeration appliance of any one of the second through fifth aspects is presented, wherein the trim breaker of the cabinet provides one or more of the interior surfaces of the cabinet defining the refrigeration compartment that the projection of the trim breaker of the door opposes when the door is in the closed position.

(44) According to a seventh aspect of the present disclosure, the refrigeration appliance of the sixth aspect is presented, wherein the one or more interior surfaces of the cabinet defining the refrigeration compartment that the trim breaker of the cabinet provides and that the projection of the trim breaker of the door opposes form part of a ceiling of the refrigeration compartment.

(45) According to an eighth aspect of the present disclosure, the refrigeration appliance of any one of the first through seventh aspects is presented, wherein the space of the door between the inner liner and the outer wrapper has an air pressure that is less than atmospheric pressure.

(46) According to a ninth aspect of the present disclosure, the refrigeration appliance of any one of the first through eighth aspects is presented, wherein (a) the trim breaker of the door further comprises (i) an outer forward recess within which an edge of the outer wrapper of the door is disposed and (ii) an inner forward recess within which an edge of the inner liner of the door is disposed, the outer forward recess disposed closer to an outer perimeter of the trim breaker than the inner forward recess, and (b) the projection extends rearward relative to the inner forward recess and into the refrigeration compartment of the cabinet.

(47) According to a tenth aspect of the present disclosure, the refrigeration appliance of the ninth aspect is presented, wherein (i) the trim breaker of the door further comprises a rear recess disposed closer to the outer perimeter than the inner forward recess but further from the outer perimeter than the outer forward recess, and (ii) a forward portion of a gasket of the door is disposed within the rear recess to couple the trim breaker and the gasket together.

(48) According to an eleventh aspect of the present disclosure, the refrigeration appliance of any one of the first through tenth aspects is presented, wherein the trim breaker of the door further comprises, at a top section thereof, (i) a top horizontal wall, (ii) a top vertical wall that extends downward from the top horizontal wall, (iii) a second horizontal wall that extends forward from the top horizontal wall, (iv) a second vertical wall that extends downward from the second horizontal wall, (v) a third horizontal wall that extends forward from the second vertical wall, (vi) a fourth horizontal wall that extends forward from the third horizontal wall, (vii) a downward transition from the third horizontal wall to the fourth horizontal wall, (viii) a third vertical wall that extends downward from the fourth horizontal wall, (ix) a fifth horizontal wall that extends rearward from the third vertical wall, (x) an angled wall that extends rearward and downward from the third vertical wall, (xi) a fourth vertical wall that extends downward from the angled wall, and (xii) a sixth horizontal wall that extends rearward from the fourth vertical wall.

(49) According to a twelfth aspect of the present disclosure, the refrigeration appliance of the eleventh aspect is presented, wherein the top horizontal wall, the top vertical wall, and the second horizontal wall define an outer forward recess within which an edge of the outer wrapper of the door is disposed, with the edge of the outer wrapper of the door disposed horizontally between the top horizontal wall and the second horizontal wall and forward of the top vertical wall.

(50) According to a thirteenth aspect of the present disclosure, the refrigeration appliance of any one of the eleventh through twelfth aspects is presented, wherein the second horizontal wall, the second vertical wall, and the third horizontal wall define a rear recess within which a forward portion of a gasket of the door is disposed to couple the trim breaker and the gasket of the door together.

(51) According to a fourteenth aspect of the present disclosure, the refrigeration appliance of any one of the eleventh through thirteenth aspects is presented, wherein the fourth horizontal wall, the third vertical wall, and the fifth horizontal wall define an inner forward recess within which an edge of the inner liner of the door is disposed.

(52) According to a fifteenth aspect of the present disclosure, the refrigeration appliance of any one of the eleventh through fourteenth aspects is presented, wherein the fourth horizontal wall, the third vertical wall, and the fifth horizontal wall define an inner forward recess within which an edge of the inner liner of the door is disposed.

(53) According to a sixteenth aspect of the present disclosure, the refrigeration appliance of any one of the eleventh through fifteenth aspects is presented, wherein the angled wall of the trim breaker of the door opposes the one or more interior surfaces of the cabinet defining the refrigeration compartment.

(54) According to a seventeenth aspect of the present disclosure, the refrigeration appliance of any

one of the first through sixteenth aspects is presented, wherein the door further comprises a gasket comprising (i) a forward portion coupled to the trim breaker and (ii) a rearward portion that provides a rear surface that is substantially vertical and positioned to interface with and form a seal against the forward surface of the outer wrapper when the door is in the closed position.

(55) According to an eighteenth aspect of the present disclosure, the refrigeration appliance of the seventeenth aspect is presented, wherein the forward portion of the gasket of the door is disposed within a rear recess of the trim breaker of the door to couple the gasket and the trim breaker together.

(56) According to a nineteenth aspect of the present disclosure, the refrigeration appliance of any one of the first through eighteenth aspects is presented, wherein the gap has a shortest distance that is less than 12 mm.

(57) According to a twentieth aspect of the present disclosure, a refrigeration appliance comprises (a) a cabinet comprising a vacuum-insulated structure within which a refrigeration compartment configured to hold a food item is disposed; and (b) a door comprising a vacuum-insulated door structure coupled to the cabinet, the vacuum-insulated door comprising (i) a gasket that forms a seal against the vacuum-insulated cabinet when the door is in a closed position and (ii) a projection inward of the gasket that projects into the refrigeration compartment and is separated from the vacuum-insulated cabinet by a gap, the gap having a shortest distance that is less than 12 mm.

(58) It will be understood by one having ordinary skill in the art that construction of the described disclosure and other components is not limited to any specific material. Other exemplary embodiments of the disclosure disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

(59) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

(60) It is also important to note that the construction and arrangement of the elements of the disclosure as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(61) It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present

disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

Claims

1. A refrigeration appliance comprising: a cabinet comprising (i) a forward surface that is substantially vertical, (ii) interior surfaces defining a refrigeration compartment configured to hold a food item, and (iii) an opening into the refrigeration compartment disposed inward of the forward surface; and a door coupled to the cabinet, the door comprising a closed position that denies access through the opening and into the refrigeration compartment of the cabinet from an external environment, an open position that allows access through the opening to the refrigeration compartment from the external environment, the door being movable to, from, and between the closed position and the open position, an inner liner facing the refrigeration compartment when the door is in the closed position, an outer wrapper separated from the inner liner by a space, and a trim breaker coupled to both the inner liner and the outer wrapper, the trim breaker comprising a projection that extends rearward into the refrigeration compartment of the cabinet, when the door is in the closed position, wherein, the projection of the trim breaker opposes one or more of the interior surfaces of the cabinet defining the refrigeration compartment, when the door is in the closed position, but is separated therefrom by a gap, and wherein, the trim breaker of the door further comprises, at a top section thereof, (i) a top horizontal wall, (ii) a top vertical wall that extends downward from the top horizontal wall, (iii) a second horizontal wall that extends forward from the top vertical wall, (iv) a second vertical wall that extends downward from the second horizontal wall, (v) a third horizontal wall that extends rearward from the second vertical wall, (vi) a fourth horizontal wall that extends rearward from the third horizontal wall, (vii) a downward transition from the third horizontal wall to the fourth horizontal wall, (viii) a third vertical wall that extends downward from the fourth horizontal wall, (ix) a fifth horizontal wall that extends forward from the third vertical wall, (x) an angled wall that extends rearward and downward from the third vertical wall, (xi) a fourth vertical wall that extends downward from the angled wall, and (xii) a sixth horizontal wall that extends forward from the fourth vertical wall.
2. The refrigeration appliance of claim 1, wherein the cabinet further comprises (i) an inner liner, (ii) an outer wrapper at least partially enveloping the inner liner of the cabinet and separated from the inner liner of the cabinet by a space, and (iii) a trim breaker coupled to both the inner liner of the cabinet and the outer wrapper of the cabinet.
3. The refrigeration appliance of claim 2, wherein the outer wrapper of the cabinet provides the forward surface, and the door further comprises a gasket that forms a seal against the forward surface of the outer wrapper of the cabinet when the door is in the closed position.
4. The refrigeration appliance of claim 2, wherein the space of the cabinet between the inner liner of the cabinet and the outer wrapper of the cabinet has an air pressure that is less than atmospheric pressure.
5. The refrigeration appliance of claim 2, wherein the trim breaker of the cabinet comprises (i) a forward recess within which an edge of the outer wrapper of the cabinet is disposed to couple the trim breaker of the cabinet and the outer wrapper of the cabinet together and (ii) a rear recess within which an edge of the inner liner of the cabinet is disposed to couple the trim breaker of the cabinet and the inner liner of the cabinet together.
6. The refrigeration appliance of claim 2, wherein the trim breaker of the cabinet provides one or more of the interior surfaces of the cabinet defining the refrigeration compartment that the projection of the trim breaker of the door opposes when the door is in the closed position.
7. The refrigeration appliance of claim 6, wherein the one or more interior surfaces of the cabinet defining the refrigeration compartment that the trim breaker of the cabinet provides and that the projection of the trim breaker of the door opposes form part of a ceiling of the refrigeration

compartment.

8. The refrigeration appliance of claim 1, wherein the space of the door between the inner liner of the door and the outer wrapper of the door has an air pressure that is less than atmospheric pressure.

9. The refrigeration appliance of claim 1, wherein the trim breaker of the door further comprises (i) an outer forward recess within which an edge of the outer wrapper of the door is disposed and (ii) an inner forward recess within which an edge of the inner liner of the door is disposed, the outer forward recess disposed closer to an outer perimeter of the trim breaker of the door than the inner forward recess, and the projection of the trim breaker of the door extends rearward relative to the inner forward recess of the trim breaker of the door and into the refrigeration compartment of the cabinet.

10. The refrigeration appliance of claim 9, wherein the trim breaker of the door further comprises a rear recess disposed closer to the outer perimeter of the trim breaker of the door than the inner forward recess of the trim breaker of the door but further from the outer perimeter of the trim breaker of the door than the outer forward recess of the trim breaker of the door, and a forward portion of a gasket of the door is disposed within the rear recess of the trim breaker of the door to couple the trim breaker of the door and the gasket together.

11. The refrigeration appliance of claim 1, wherein the top horizontal wall, the top vertical wall, and the second horizontal wall of the trim breaker of the door define an outer forward recess within which an edge of the outer wrapper of the door is disposed, with the edge of the outer wrapper of the door disposed horizontally between the top horizontal wall of the trim breaker of the door and the second horizontal wall of the trim breaker of the door and forward of the top vertical wall of the trim breaker of the door.

12. The refrigeration appliance of claim 1, wherein the second horizontal wall, the second vertical wall, and the third horizontal wall of the trim breaker of the door define a rear recess within which a forward portion of a gasket of the door is disposed to couple the trim breaker of the door and the gasket of the door together.

13. The refrigeration appliance of claim 1, wherein the fourth horizontal wall, the third vertical wall, and the fifth horizontal wall of the trim breaker of the door define an inner forward recess within which an edge of the inner liner of the door is disposed.

14. The refrigeration appliance of claim 13, wherein the inner forward recess of the trim breaker of the door is disposed elevationally higher than the angled wall of the trim breaker of the door.

15. The refrigeration appliance of claim 1, wherein the angled wall, the fourth vertical wall, and the sixth horizontal wall of the trim breaker of the door define the projection.

16. The refrigeration appliance of claim 1, wherein the angled wall of the trim breaker of the door opposes the one or more interior surfaces of the cabinet defining the refrigeration compartment.

17. The refrigeration appliance of claim 1, wherein the door further comprises a gasket comprising (i) a forward portion coupled to the trim breaker of the door and (ii) a rearward portion that provides a rear surface that is substantially vertical and positioned to interface with and form a seal against the forward surface of the outer wrapper of the cabinet when the door is in the closed position.

18. The refrigeration appliance of claim 17, wherein the forward portion of the gasket of the door is disposed within a rear recess of the trim breaker of the door to couple the gasket of the door and the trim breaker of the door together.

19. The refrigeration appliance of claim 1, wherein the gap has a shortest distance that is less than 12 mm.

20. The refrigeration appliance of claim 13, wherein the top horizontal wall, the top vertical wall, and the second horizontal wall of the trim breaker of the door define an outer forward recess within which an edge of the outer wrapper of the door is disposed, with the edge of the outer wrapper of the door disposed horizontally between the top horizontal wall and the second horizontal wall of

the trim breaker of the door and forward of the top vertical wall of the trim breaker of the door, the angled wall, the fourth vertical wall, and the sixth horizontal wall of the trim breaker of the door define the projection, and the inner forward recess of the trim breaker of the door is elevationally between the outer forward recess of the trim breaker of the door and the angled wall of the trim breaker of the door.
